

Equity Dilution Protection and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Dilution Protection (EDP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on EDP achieves an annualized gross (net) Sharpe ratio of 0.53 (0.42), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (17) bps/month with a t-statistic of 3.07 (2.47), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Change in financial liabilities, Net external financing, Asset growth, Inventory Growth, change in net operating assets) is 19 bps/month with a t-statistic of 2.75.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between current and future consumption, linking cross-sectional variation in expected returns to firms' differential exposures to systematic consumption risk. While consumption-based asset pricing models provide a theoretical foundation for understanding risk premia, empirical tests often struggle to identify firm characteristics that reliably capture variation in consumption risk exposure across assets. We identify equity dilution protection (EDP) as a novel predictor of cross-sectional stock returns that relates to firms' sensitivity to aggregate consumption dynamics. Firms that avoid equity dilution during periods of high marginal utility preserve shareholder value when consumption growth is low, providing insurance against adverse consumption states that rational investors value through lower required returns.

The consumption-based asset pricing framework establishes that expected returns compensate investors for bearing systematic consumption risk, with assets that co-vary positively with the stochastic discount factor commanding higher risk premia (Lucas Jr, 1978; Breeden, 1979). Equity dilution protection captures firms' ability to fund operations and growth without issuing new shares, thereby avoiding wealth transfers from existing shareholders during states of high marginal utility. When consumption growth slows and marginal utility rises, firms with stronger dilution protection provide relatively stable cash flows to investors, reducing their exposure to consumption risk (Campbell and Cochrane, 1999). This protective characteristic becomes particularly valuable during disaster states or periods of heightened macro uncertainty, when the marginal utility of consumption peaks and investors demand higher compensation for bearing additional risk (Barro, 2006; Gabaix, 2012). The state-dependent nature of dilution protection implies that firms maintaining higher EDP exhibit lower sensitivity to long-run consumption risks. Under habit formation preferences, these firms' cash flows correlate less strongly with the surplus consump-

tion ratio, reducing their systematic risk exposure (Campbell and Cochrane, 1999). During bad times characterized by low consumption growth or high consumption volatility, firms with strong dilution protection effectively decouple their shareholders' wealth from adverse consumption shocks, lowering the covariance between their returns and the intertemporal marginal rate of substitution. The consumption risk channel predicts that high EDP firms command lower expected returns in equilibrium. Rational investors recognize that these firms provide consumption hedging benefits during states when marginal utility is high, accepting lower average returns as compensation for this insurance (Merton, 1973). This mechanism operates through both direct and indirect channels: directly through reduced dilution during high marginal utility states, and indirectly through these firms' enhanced financial flexibility that allows them to maintain stable operations when consumption growth deteriorates (Bansal and Yaron, 2004).

We find strong evidence that equity dilution protection predicts cross-sectional variation in stock returns. A value-weighted long/short trading strategy based on EDP achieves an annualized gross Sharpe ratio of 0.53, with a monthly average excess return of 27 basis points and a t-statistic of 3.81. The strategy's performance remains robust across multiple factor model specifications, generating a monthly alpha of 22 basis points relative to the Fama-French six-factor model with a t-statistic of 3.07. The economic magnitude of the EDP effect is substantial and pervasive across different segments of the market. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the EDP strategy delivers an average return of 33 basis points per month with a t-statistic of 3.81, demonstrating that the anomaly is not confined to small, illiquid stocks where trading costs might prohibit implementation. The strategy's alpha relative to the six-factor model among large-cap stocks equals 26 basis points per month with a t-statistic of 2.96, confirming that the return predictability cannot be explained by known risk factors. After

accounting for transaction costs using high-frequency bid-ask spread estimates, the EDP strategy maintains economic significance with a net Sharpe ratio of 0.42. The strategy generates positive net generalized alphas relative to all five standard factor models, with the net alpha relative to the six-factor model equaling 17 basis points per month and a t-statistic of 2.47. Controlling for the six most closely related anomalies from the factor zoo plus the six Fama-French factors, the EDP strategy continues to generate an alpha of 19 basis points per month with a t-statistic of 2.75, indicating that the signal contains unique information about expected returns beyond existing predictors.

Our study contributes to the consumption-based asset pricing literature by identifying equity dilution protection as a characteristic that captures cross-sectional variation in consumption risk exposure. While [Lettau and Ludvigson \(2001\)](#) and [Parker and Julliard \(2005\)](#) document the importance of consumption risk for aggregate stock returns, we demonstrate that firm-level dilution decisions create heterogeneous exposures to consumption states that generate predictable return variation. Our findings extend [Jagannathan and Wang \(2016\)](#)'s analysis of consumption-based factors by showing that financing choices, specifically equity issuance decisions, serve as observable proxies for otherwise difficult-to-measure consumption risk loadings. We advance the literature on equity financing and returns by providing a consumption-based interpretation for the observed return predictability. Previous work by [Daniel and Titman \(2006\)](#) and [Pontiff and Woodgate \(2008\)](#) documents that equity issuance predicts negative returns but primarily attributes this pattern to mispricing or agency conflicts. Our consumption risk framework offers a rational explanation: firms that protect against dilution provide consumption hedging benefits that lower their required returns in equilibrium, consistent with the predictions of [Merton \(1973\)](#) regarding assets' hedging characteristics and expected returns. Methodologically, we employ the rigorous testing protocol of [Novy-Marx and Velikov \(2023b\)](#) to establish

the robustness of the EDP anomaly, addressing concerns about data mining and spurious discoveries in the factor zoo. The EDP signal’s gross Sharpe ratio exceeds 92% of documented anomalies, while its net Sharpe ratio after transaction costs surpasses 96% of existing strategies, positioning it among the most robust cross-sectional return predictors. These findings have important implications for portfolio construction and risk management. The consumption risk interpretation suggests that the EDP premium represents compensation for bearing systematic risk rather than an arbitrage opportunity, implying that the return predictability should persist even as it becomes widely known. Furthermore, the state-dependent nature of the premium—stronger during high marginal utility states—provides guidance for dynamic portfolio strategies that account for time-varying consumption risk exposures.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms’ financing activities. construction of our Equity Dilution Protection signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities with maturities exceeding one year. Stockholders’ equity (SEQ) measures the total common and preferred stockholders’ equity, representing the book value of shareholders’ residual claim on the firm’s assets after deducting liabilities. This variable serves as a natural scaling factor that adjusts for firm size and provides economic context for the magnitude of debt issuance activities. construct the Equity Dilution Protection signal by computing

the year-over-year change in long-term debt issuance scaled by lagged stockholders' equity: $(DLTIS(t) - DLTIS(t-1)) / SEQ(t-1)$. This measure captures the change in a firm's reliance on debt financing relative to its equity base. A positive value indicates an acceleration in debt issuance activity, suggesting that the firm increasingly turns to debt markets rather than equity markets for external financing, potentially protecting existing shareholders from dilution. We calculate this signal using fiscal year-end values and require non-missing values for both DLTIS and SEQ in consecutive years. To ensure meaningful economic interpretation, we exclude observations where lagged stockholders' equity is negative or zero, as these cases represent firms in financial distress where the scaling would produce unreliable metrics.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDP signal. Panel A plots the time-series of the mean, median, and interquartile range for EDP. On average, the cross-sectional mean (median) EDP is -0.29 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input EDP data. The signal's interquartile range spans -0.17 to 0.16. Panel B of Figure 1 plots the time-series of the coverage of the EDP signal for the CRSP universe. On average, the EDP signal is available for 6.24% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does EDP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDP portfolio and sells the low EDP portfolio. The rest

of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short EDP strategy earns an average return of 0.27% per month with a t-statistic of 3.81. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.22% to 0.33% per month and have t-statistics exceeding 3.07 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.30, with a t-statistic of 6.35 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 551 stocks and an average market capitalization of at least \$1,439 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each

portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 23 bps/month with a t-statistics of 4.86. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -3-24bps/month. The lowest return, (-3 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.48. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the EDP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDP, as well as average returns and alphas for long/short trading EDP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EDP strategy achieves an average return of 33 bps/month with a t-statistic of 3.81. Among these large cap stocks, the alphas for the EDP strat-

egy relative to the five most common factor models range from 26 to 39 bps/month with t-statistics between 2.96 and 4.38.

5 How does EDP perform relative to the zoo?

Figure 2 puts the performance of EDP in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EDP strategy falls in the distribution. The EDP strategy’s gross (net) Sharpe ratio of 0.53 (0.42) is greater than 92% (96%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EDP strategy (red line).² Ignoring trading costs, a \$1 invested in the EDP strategy would have yielded \$3.40 which ranks the EDP strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDP strategy would have yielded \$2.09 which ranks the EDP strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDP relative to those. Panel A shows that the EDP strategy gross alphas fall between the 64 and 72 percentiles across the five

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDP strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDP ranks between the 78 and 85 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EDP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDP with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDP or at least to weaken the power EDP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDP signal in these Fama-MacBeth regressions exceed 1.16, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on EDP is 0.03.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDP strategy earns alphas that range from 18-21bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.67, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDP trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.75.

7 Does EDP add relative to the whole zoo?

Finally, we can ask how much adding EDP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the EDP signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes EDP grows to \$1183.39.

8 Conclusion

This study provides robust evidence that Equity Dilution Protection (EDP) represents a significant and economically meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDP-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on EDP delivers an annualized gross Sharpe ratio of 0.53, which remains economically signifi-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDP is available.

cant at 0.42 after accounting for transaction costs. The strategy’s ability to generate monthly abnormal returns of 22 basis points gross (17 basis points net) relative to the Fama-French five-factor model plus momentum, with t-statistics exceeding conventional significance thresholds (3.07 and 2.47, respectively), underscores its statistical reliability. Particularly noteworthy is the signal’s robustness when tested against a comprehensive set of twelve factors, including the six most closely related strategies from the factor zoo. The persistence of a 19 basis points monthly alpha with a t-statistic of 2.75 suggests that EDP captures unique information about future returns not subsumed by existing factors related to financing activities, asset growth, or operating changes.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The net Sharpe ratio of 0.42 indicates that the strategy remains profitable after realistic implementation costs, suggesting potential real-world applicability. The signal’s distinctiveness from related factors such as net debt financing, changes in financial liabilities, and net external financing indicates that monitoring equity dilution protection provides incremental value in portfolio construction and risk management.

However, several limitations warrant consideration. First, while our analysis accounts for transaction costs, other implementation challenges such as market impact, especially for less liquid stocks, may further erode returns in practice. Second, the study period and market conditions examined may not fully capture the signal’s performance across different market regimes or economic cycles. Third, the underlying economic mechanisms driving the EDP anomaly require further theoretical development to understand whether this represents a risk-based explanation or a behavioral mispricing that may be arbitrated away as it becomes more widely known.

Future research should explore several promising avenues. Investigation into the time-varying nature of the EDP premium could reveal whether the signal’s effective-

ness varies with market conditions, regulatory changes, or macroeconomic factors. Cross-country and cross-asset class studies would help establish the generalizability of these findings beyond the current sample. Additionally, examining the interaction between EDP and corporate governance characteristics, institutional ownership, or analyst coverage could provide insights into the economic channels through which equity dilution protection affects returns. Finally, developing a deeper theoretical framework linking equity dilution protection to firm value and expected returns would enhance our understanding of why this signal possesses predictive power and under what conditions it might weaken or strengthen.

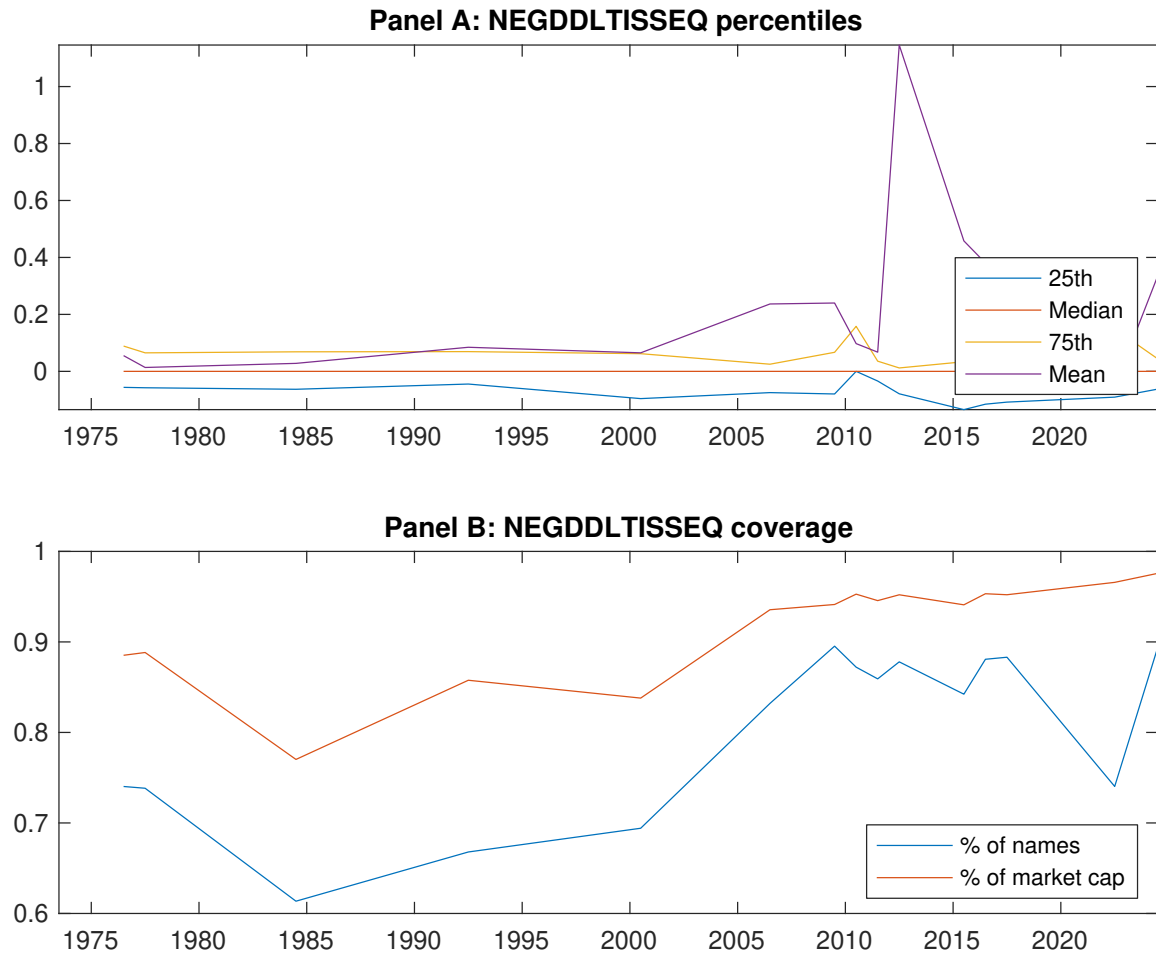


Figure 1: Times series of EDP percentiles and coverage.
This figure plots descriptive statistics for EDP. Panel A shows cross-sectional percentiles of EDP over the sample. Panel B plots the monthly coverage of EDP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on EDP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55 [2.57]	0.66 [3.66]	0.64 [3.24]	0.73 [4.10]	0.82 [4.15]	0.27 [3.81]
α_{CAPM}	-0.18 [-3.14]	0.05 [0.98]	-0.02 [-0.41]	0.12 [2.67]	0.15 [2.82]	0.33 [4.68]
α_{FF3}	-0.20 [-3.57]	0.00 [0.06]	0.04 [0.81]	0.11 [2.37]	0.13 [2.45]	0.33 [4.66]
α_{FF4}	-0.17 [-2.97]	0.02 [0.42]	0.07 [1.33]	0.08 [1.74]	0.11 [2.14]	0.28 [3.97]
α_{FF5}	-0.20 [-3.49]	-0.05 [-1.06]	0.10 [1.87]	0.03 [0.70]	0.05 [0.92]	0.24 [3.48]
α_{FF6}	-0.17 [-3.08]	-0.03 [-0.71]	0.12 [2.19]	0.02 [0.36]	0.04 [0.82]	0.22 [3.07]
Panel B: Fama and French (2018) 6-factor model loadings for EDP-sorted portfolios						
β_{MKT}	1.09 [83.34]	0.98 [95.31]	0.97 [75.45]	0.97 [90.95]	1.04 [86.06]	-0.05 [-2.98]
β_{SMB}	0.11 [5.68]	-0.11 [-7.02]	-0.03 [-1.35]	-0.03 [-2.12]	0.13 [7.21]	0.02 [0.79]
β_{HML}	0.08 [3.04]	0.14 [7.11]	-0.14 [-5.80]	-0.02 [-0.74]	-0.05 [-2.05]	-0.12 [-3.95]
β_{RMW}	0.10 [3.75]	0.12 [5.94]	-0.08 [-3.01]	0.08 [4.06]	0.12 [4.98]	0.02 [0.69]
β_{CMA}	-0.14 [-3.60]	0.03 [1.01]	-0.11 [-3.06]	0.18 [5.74]	0.17 [4.68]	0.30 [6.35]
β_{UMD}	-0.04 [-2.76]	-0.02 [-2.40]	-0.03 [-2.28]	0.02 [2.33]	0.01 [0.58]	0.04 [2.64]
Panel C: Average number of firms (n) and market capitalization (me)						
n	649	551	1069	605	621	
me (\$10 ⁶)	1495	3003	2305	3111	1439	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [3.81]	0.33 [4.68]	0.33 [4.66]	0.28 [3.97]	0.24 [3.48]	0.22 [3.07]
Quintile	NYSE	EW	0.23 [4.86]	0.25 [5.44]	0.24 [5.19]	0.23 [4.74]	0.21 [4.54]	0.20 [4.29]
Quintile	Name	VW	0.26 [3.68]	0.32 [4.74]	0.33 [4.78]	0.27 [3.98]	0.27 [3.88]	0.23 [3.37]
Quintile	Cap	VW	0.24 [3.65]	0.29 [4.37]	0.29 [4.39]	0.23 [3.47]	0.20 [3.11]	0.16 [2.52]
Decile	NYSE	VW	0.31 [3.23]	0.38 [3.99]	0.38 [4.01]	0.33 [3.42]	0.30 [3.09]	0.27 [2.75]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [2.99]	0.26 [3.70]	0.26 [3.71]	0.23 [3.35]	0.19 [2.74]	0.17 [2.47]
Quintile	NYSE	EW	-0.03 [-0.48]					
Quintile	Name	VW	0.20 [2.83]	0.26 [3.80]	0.26 [3.80]	0.23 [3.38]	0.21 [3.08]	0.18 [2.76]
Quintile	Cap	VW	0.19 [2.89]	0.24 [3.64]	0.24 [3.64]	0.21 [3.17]	0.17 [2.60]	0.14 [2.23]
Decile	NYSE	VW	0.24 [2.50]	0.30 [3.18]	0.31 [3.19]	0.28 [2.89]	0.24 [2.46]	0.21 [2.22]

Table 3: Conditional sort on size and EDP

This table presents results for conditional double sorts on size and EDP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDP and short stocks with low EDP. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EDP Quintiles					EDP Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.66 [2.39]	0.87 [3.20]	0.97 [3.58]	0.85 [3.06]	0.80 [2.87]	0.14 [1.49]	0.17 [1.91]	0.16 [1.78]	0.11 [1.14]	0.09 [0.99]	0.06 [0.61]
	(2)	0.71 [2.66]	0.91 [3.61]	0.83 [3.32]	0.86 [3.52]	0.87 [3.43]	0.16 [2.14]	0.20 [2.57]	0.16 [2.21]	0.17 [2.29]	0.12 [1.64]	0.14 [1.77]
	(3)	0.79 [3.16]	0.78 [3.55]	0.85 [3.55]	0.80 [3.52]	0.90 [3.82]	0.11 [1.40]	0.16 [2.02]	0.16 [2.02]	0.12 [1.52]	0.15 [1.83]	0.12 [1.50]
	(4)	0.70 [3.03]	0.81 [3.95]	0.86 [3.88]	0.70 [3.39]	0.91 [4.19]	0.21 [2.88]	0.25 [3.43]	0.24 [3.23]	0.21 [2.75]	0.21 [2.81]	0.19 [2.47]
	(5)	0.45 [2.26]	0.62 [3.54]	0.59 [2.97]	0.65 [3.62]	0.79 [4.12]	0.33 [3.81]	0.37 [4.19]	0.39 [4.38]	0.32 [3.63]	0.31 [3.43]	0.26 [2.96]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EDP Quintiles					EDP Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	391	394	393	393	389	37	34	32	33	35	
	(2)	107	107	107	108	107	61	62	60	61	61	
	(3)	76	77	76	77	76	108	109	105	106	108	
	(4)	64	65	64	65	64	230	238	230	234	231	
(5)	59	59	59	59	59	1384	2158	1902	2148	1585		

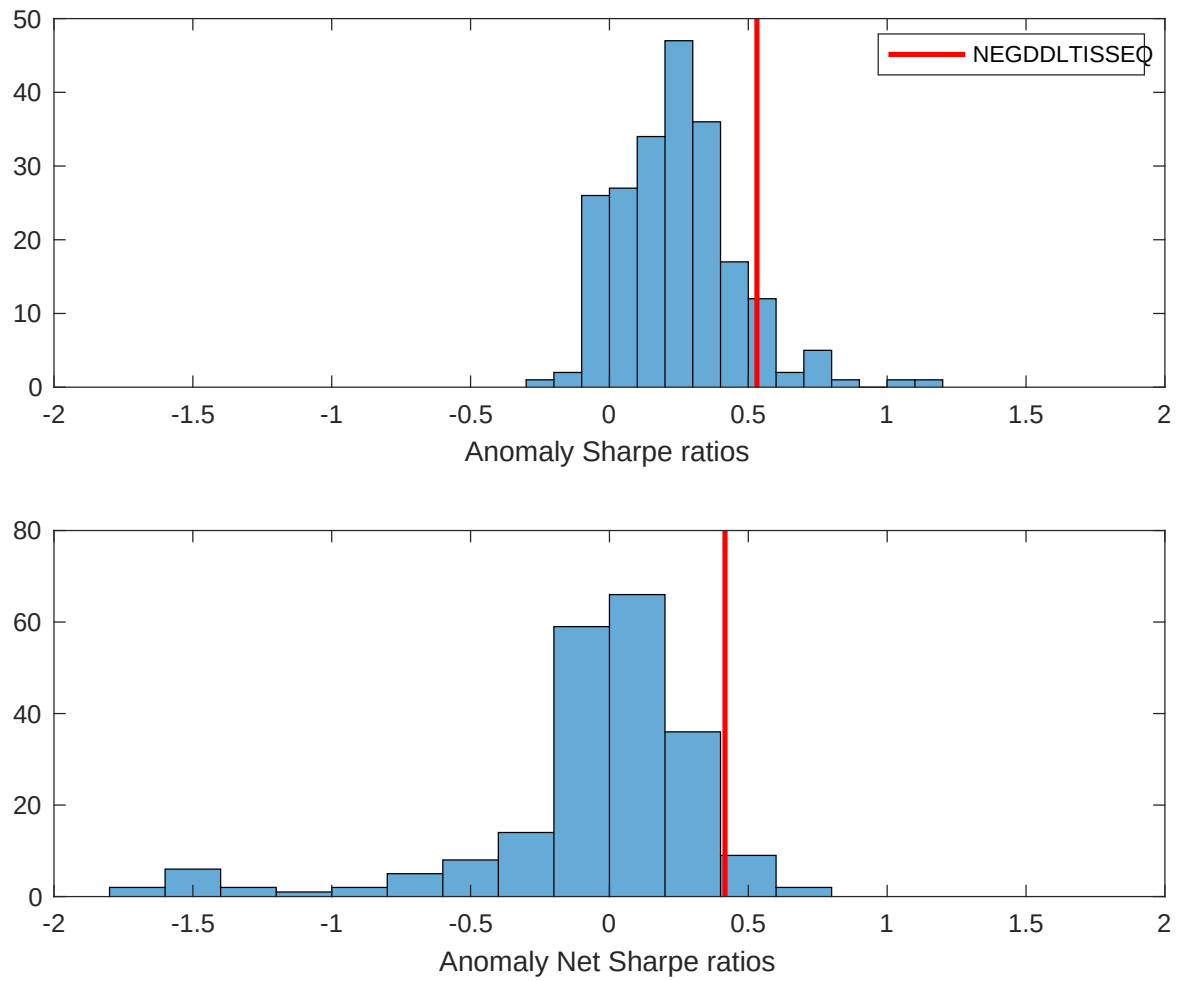


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

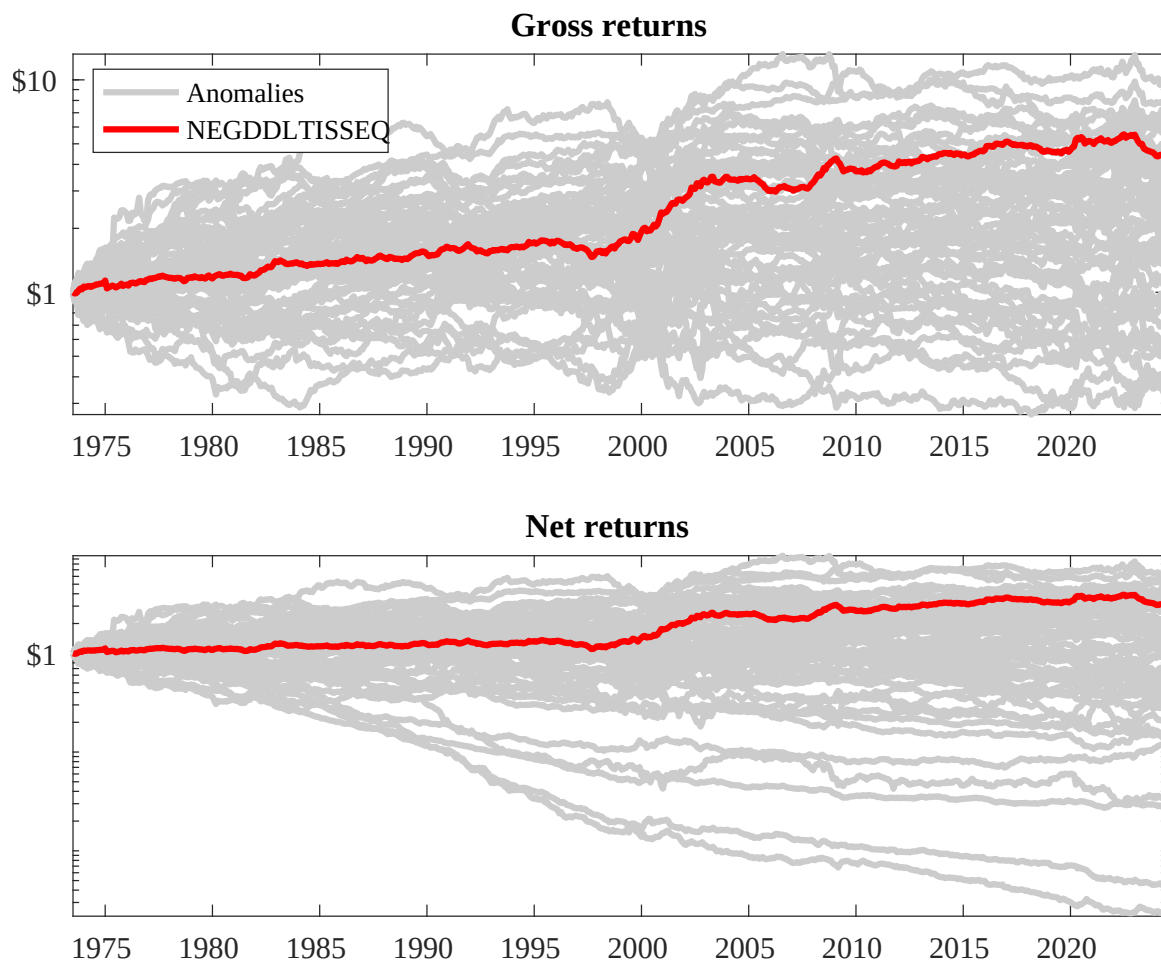
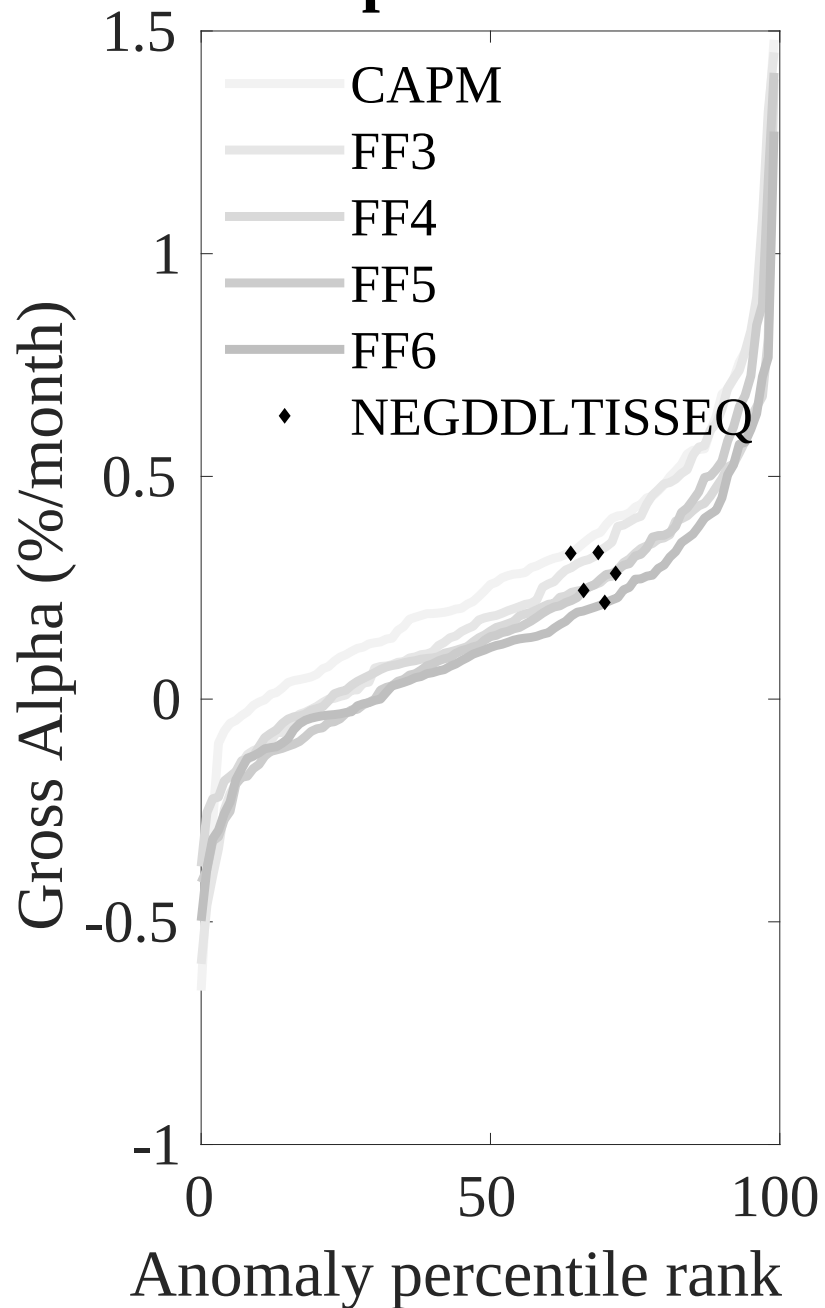


Figure 3: Dollar invested.

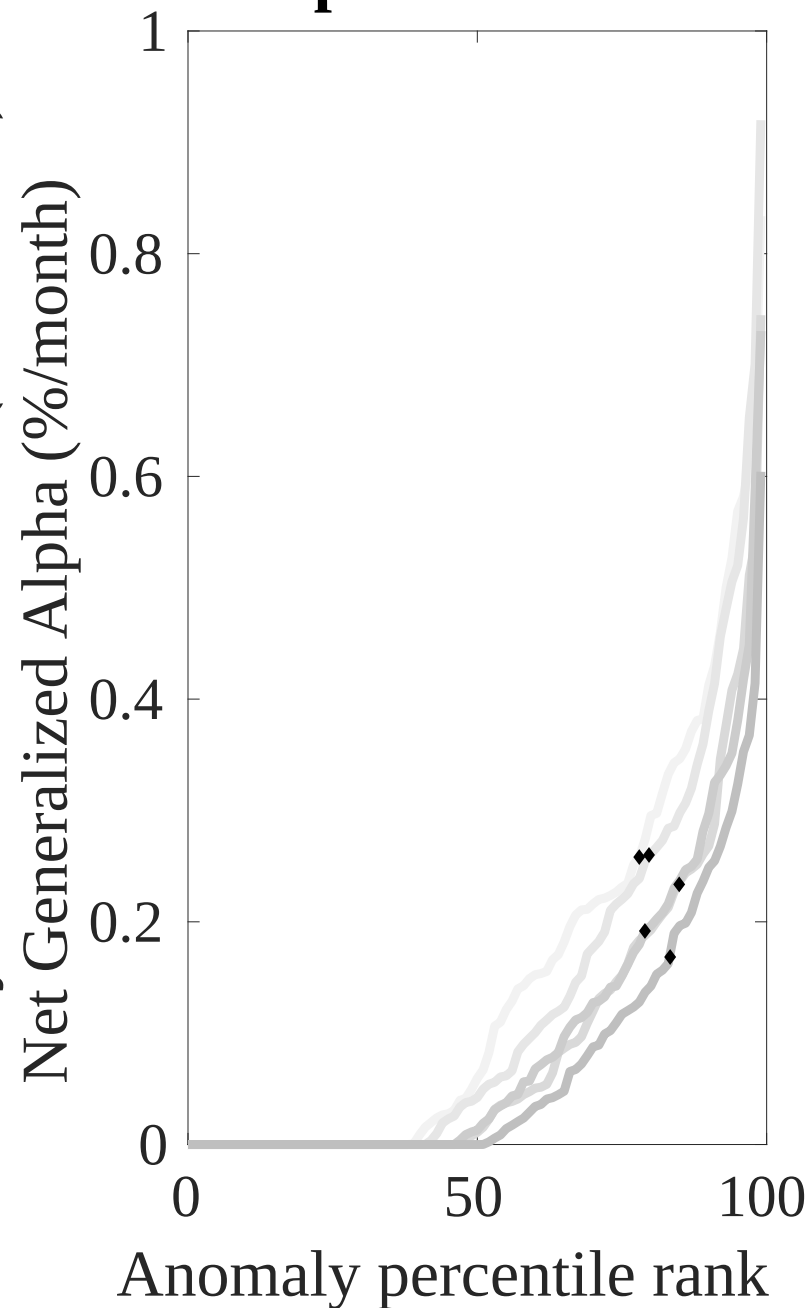
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



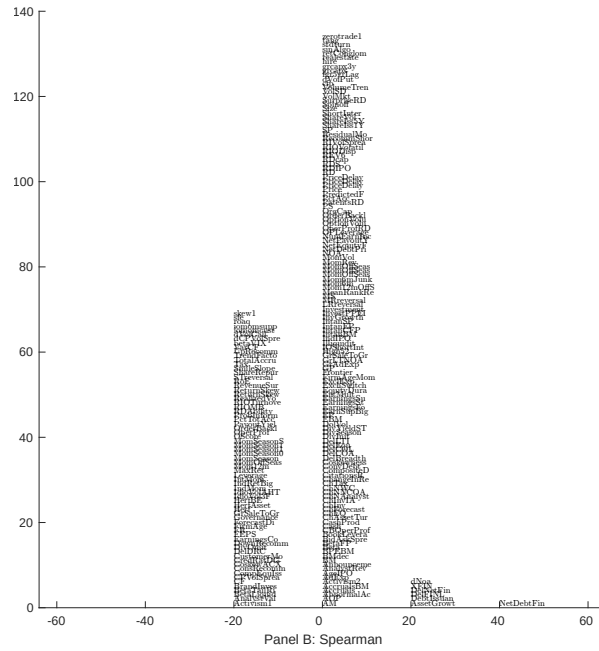
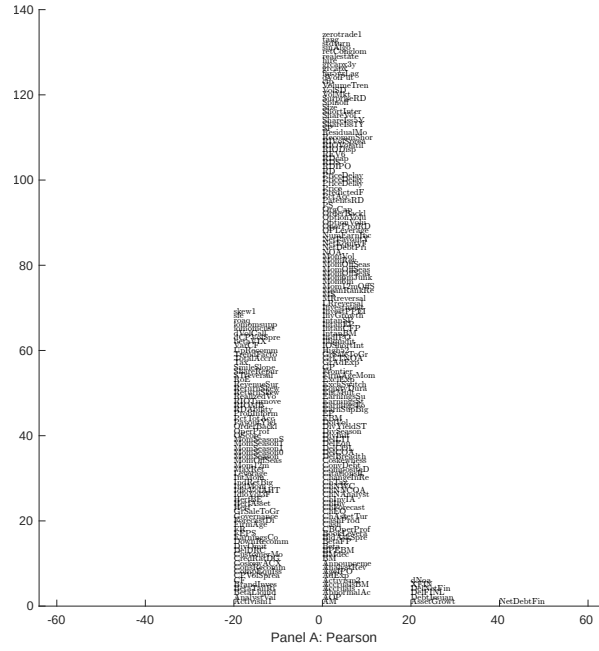


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

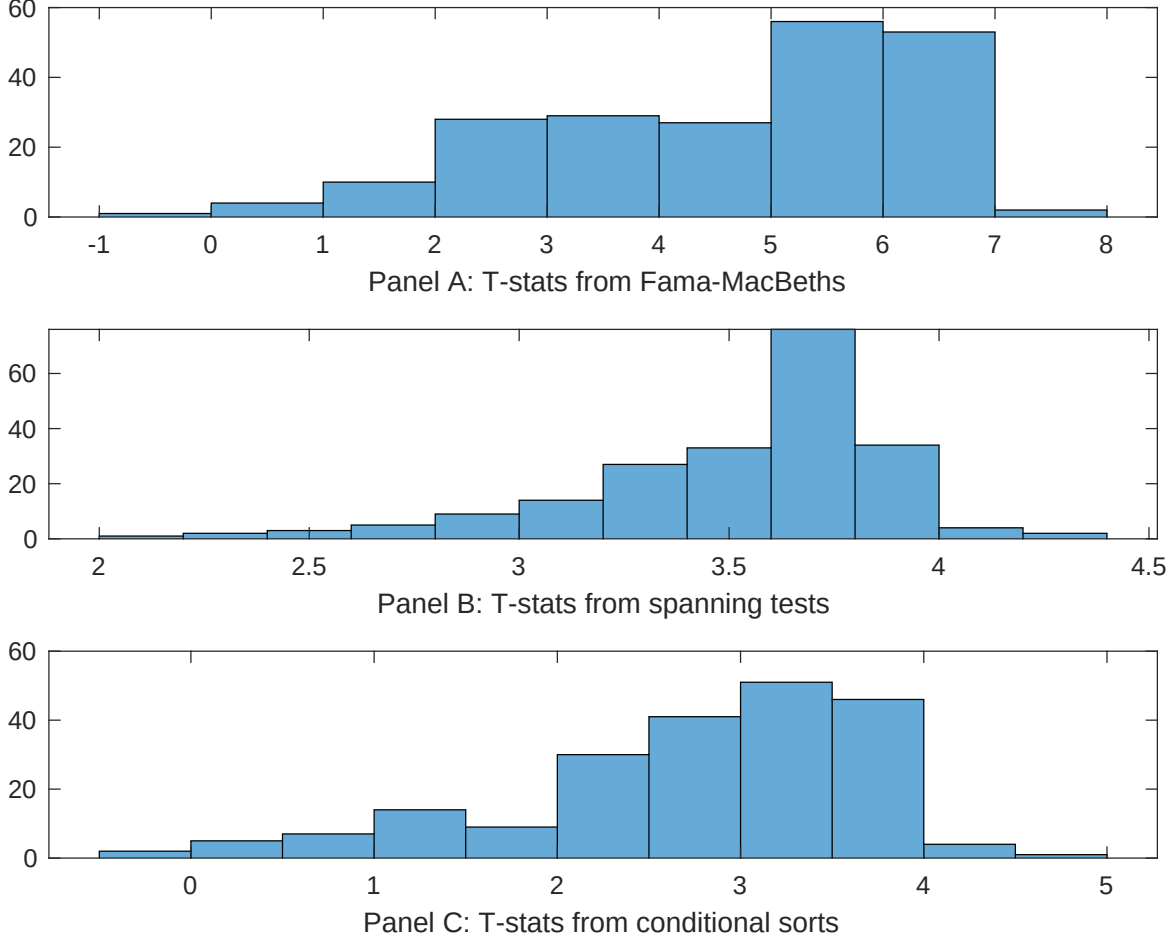


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDP}EDP_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.32]	0.13 [5.36]	0.14 [5.68]	0.14 [5.79]	0.13 [5.30]	0.14 [5.65]	0.15 [5.69]
EDP	0.56 [2.01]	0.45 [1.71]	0.78 [2.62]	0.36 [1.29]	0.12 [4.08]	0.30 [1.16]	0.10 [0.03]
Anomaly 1	0.20 [8.86]						0.98 [1.47]
Anomaly 2		0.17 [9.40]					-0.11 [-2.24]
Anomaly 3			0.19 [6.33]				0.10 [1.73]
Anomaly 4				0.10 [8.63]			0.40 [1.76]
Anomaly 5					0.42 [7.10]		0.45 [0.81]
Anomaly 6						0.14 [9.84]	0.94 [5.23]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EDP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDP} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.18 [2.73]	0.18 [2.67]	0.19 [2.70]	0.20 [2.88]	0.21 [2.99]	0.19 [2.79]	0.19 [2.75]
Anomaly 1	20.49 [5.51]						13.34 [2.62]
Anomaly 2		17.00 [4.38]					8.55 [1.61]
Anomaly 3			12.49 [3.62]				7.78 [2.12]
Anomaly 4				7.65 [1.90]			3.60 [0.84]
Anomaly 5					7.63 [2.83]		7.13 [2.61]
Anomaly 6						4.28 [1.08]	-6.52 [-1.47]
mkt	-4.70 [-3.01]	-4.35 [-2.76]	-3.03 [-1.83]	-4.75 [-2.98]	-4.88 [-3.07]	-4.73 [-2.96]	-3.60 [-2.20]
smb	-1.23 [-0.51]	-1.16 [-0.48]	4.60 [1.74]	0.03 [0.01]	1.42 [0.59]	0.66 [0.27]	1.43 [0.52]
hml	-12.62 [-4.19]	-12.32 [-4.04]	-12.06 [-3.92]	-13.80 [-4.47]	-13.69 [-4.46]	-13.80 [-4.45]	-11.34 [-3.70]
rmw	-0.11 [-0.04]	0.61 [0.20]	-5.32 [-1.43]	2.17 [0.69]	3.25 [1.03]	2.21 [0.71]	-3.81 [-1.02]
cma	26.84 [5.74]	26.47 [5.52]	24.01 [4.66]	22.89 [3.40]	25.38 [4.85]	28.99 [5.26]	14.91 [2.17]
umd	3.08 [1.96]	2.93 [1.83]	4.32 [2.74]	4.57 [2.87]	3.62 [2.26]	4.27 [2.67]	2.30 [1.43]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	18	16	15	14	15	14	19

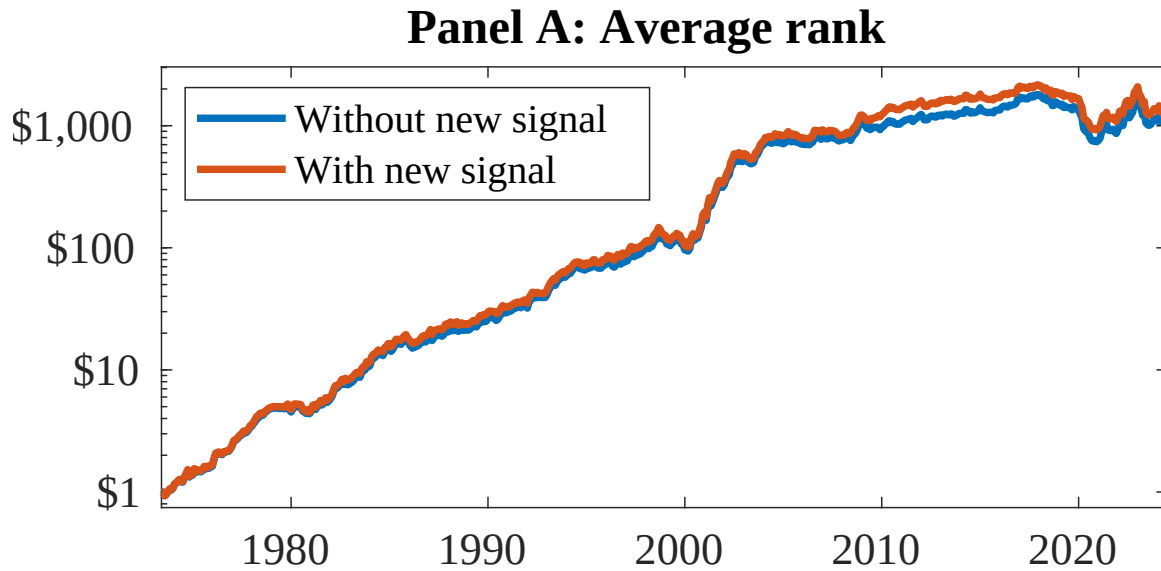


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as EDP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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